

Cellular Automata in Traffic Modeling

Cellular automata (CA) are discrete models consisting of an array or grid of cells. The cells can be arranged in any number of dimensions, but are typically arranged in 1D or 2D. Each cell within the system can be in one of a finite number of states. A group of cells, typically adjacent cells, is referred to as a neighborhood. The model creates a new generation of cells, incrementing the timestep, based on a set of rules. These rules tend to take into account the state of the current cell and the states of its neighboring cells.

A popular example of a cellular automaton is Conway's Game of Life. This program takes place on a 2D grid of infinite width and height, with each cell representing one of two states: live or dead. The rules for each generation of the game are the following:

- Any live cell with fewer than two live neighbors dies.
- Any live cell with two or three live neighbors keeps living.
- Any live cell with more than three live neighbors dies.
- Any dead cell with exactly three live neighbors becomes alive.

Despite the simplicity of these rules, they can produce complex patterns and behaviors over many generations. This demonstrates how cellular automata can model complex systems through relatively simple interactions between neighboring cells.

The simple rules used in cellular automata can produce complex behaviors, making CA useful for modeling real-world systems. One application of cellular automata is traffic modeling, where cells can represent sections of a roadway and their states can represent the presence or movement of vehicles. By defining rules that control how vehicles interact and move between cells, cellular automata can be used to simulate traffic flow and study behaviors such as congestion and traffic disruptions. This application of cellular automata provides the focus for the following literature review.